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# CRRT

**Medical Overview**  
**May. 2016**

# Continuous Renal Replacement Therapy

Also known as “slow Continuous Renal Replacement Therapy”.



“Any extracorporeal blood purification therapy intended to substitute for impaired renal function over an extended period of time and applied for or aimed at being applied for 24 hours/day.”

R. Bellomo, C Ronco and R. Mehta,  
Nomenclature for Continuous Renal  
Replacement Therapies, AJKD, Vol 28,  
November 1996

# Why Choose CRRT?

## CRRT closely mimics the native kidney.

- Slow, gentle and continuous reduce risk for hypotension
- Well tolerated by hemodynamically unstable patients
- Prevent further damage to kidney tissue
- Promote healing and renal recovery
- Regulates electrolytes, acid-base balance
- Removes large amounts of fluid and waste products over time



# CRRT is

## Groups of

### Group of words

- Continuous
- Renal
- Replacement
- Therapy

### Through group of Mechanisms

- Ultrafiltration
- Diffusion
- Convection
- Adsorption

### Describes group of treatment modalities

- SCUF
- CVVHD
- CVVH
- CVVHDF

# Molecular Transport Mechanisms

Ultrafiltration

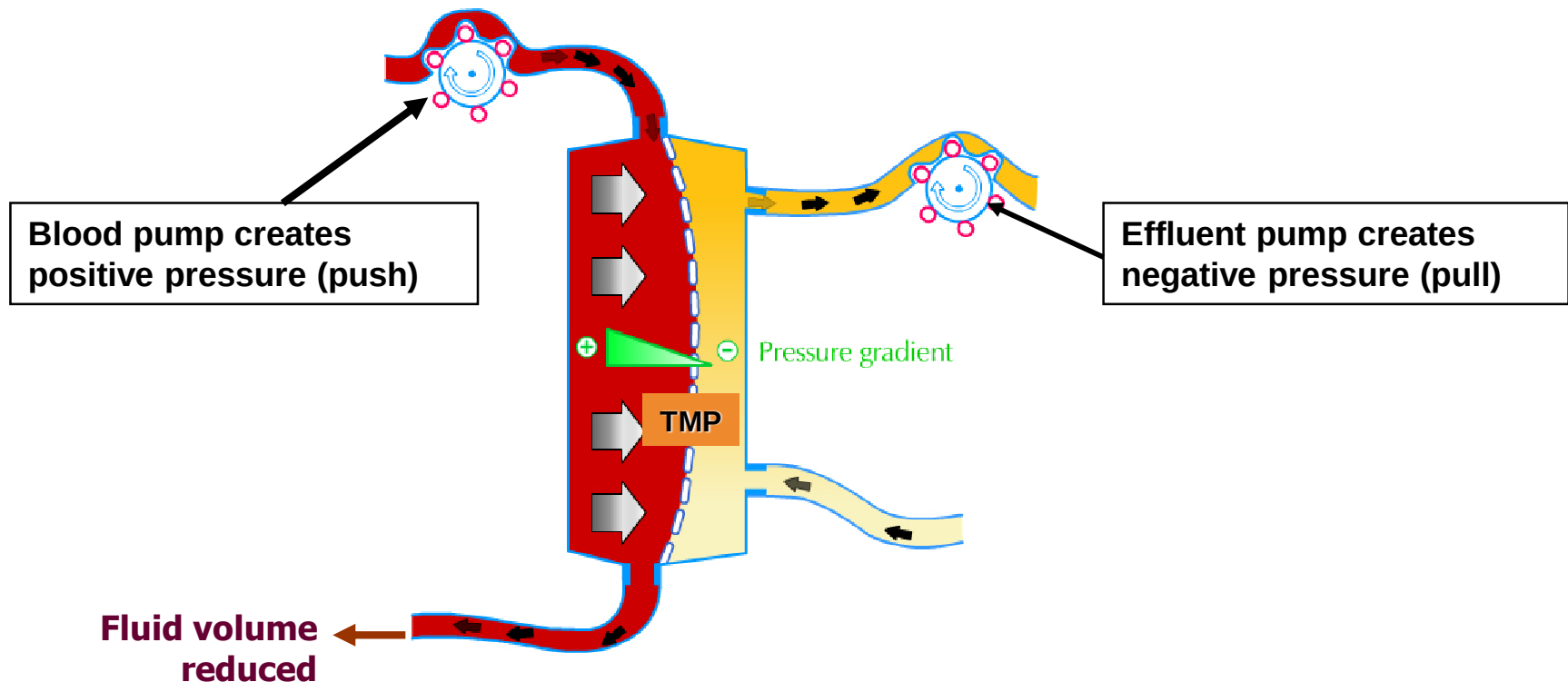
Diffusion

Convection

Adsorption

# Ultrafiltration

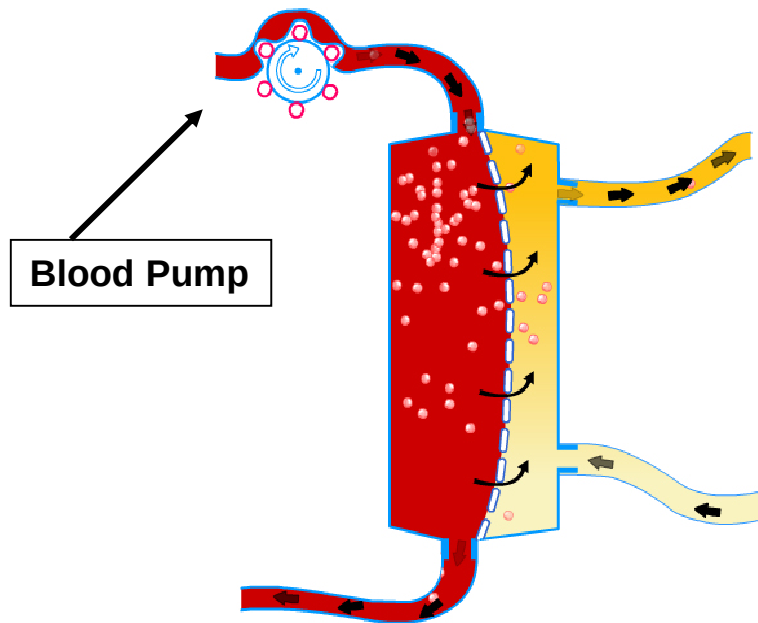
The movement of fluid through a semi-permeable membrane driven by a pressure gradient.



The creation of a pressure gradient forces plasma water across the semi-permeable membrane.

# Diffusion

Movement of solutes from an area of higher to lower solute concentration.



## Factors affecting diffusion :

Concentration gradient

Blood flow rate ( $Q_B$ )

Dialysis fluid flow rate ( $Q_D$ )

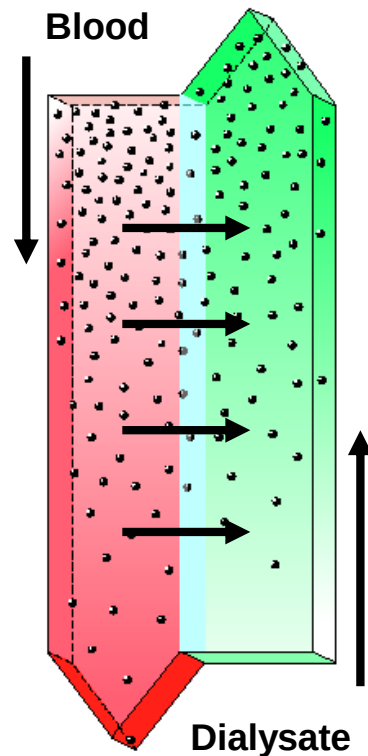
Molecular size

Filter characteristics

- Membrane type- thickness, surface area

# Dialysate Flow and Diffusive Clearance

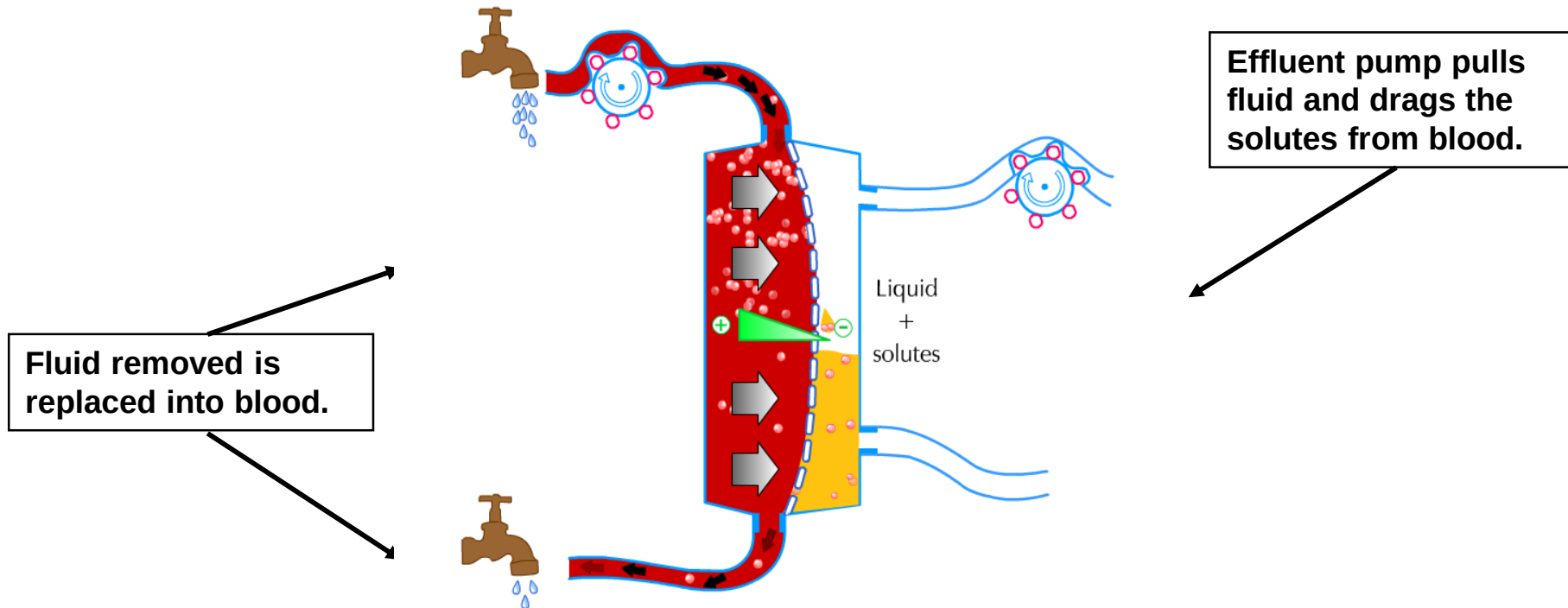
Dialysate Flows counter current to blood flow.



Counter current flow for effective diffusion  
Assures efficient solute transport of  
metabolic waste from blood to dialysate



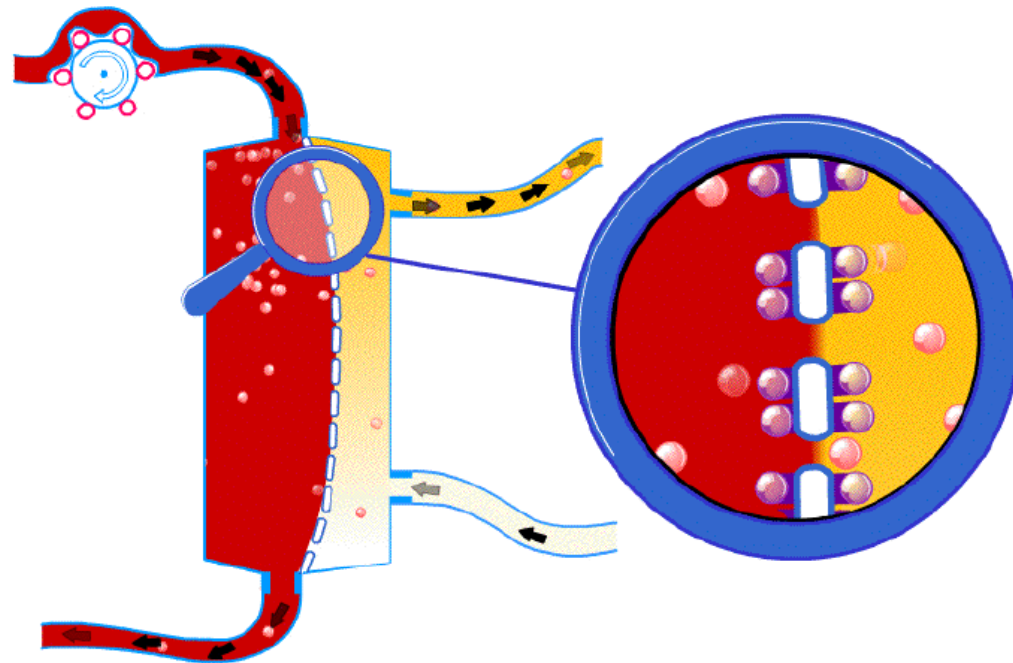
# Convection



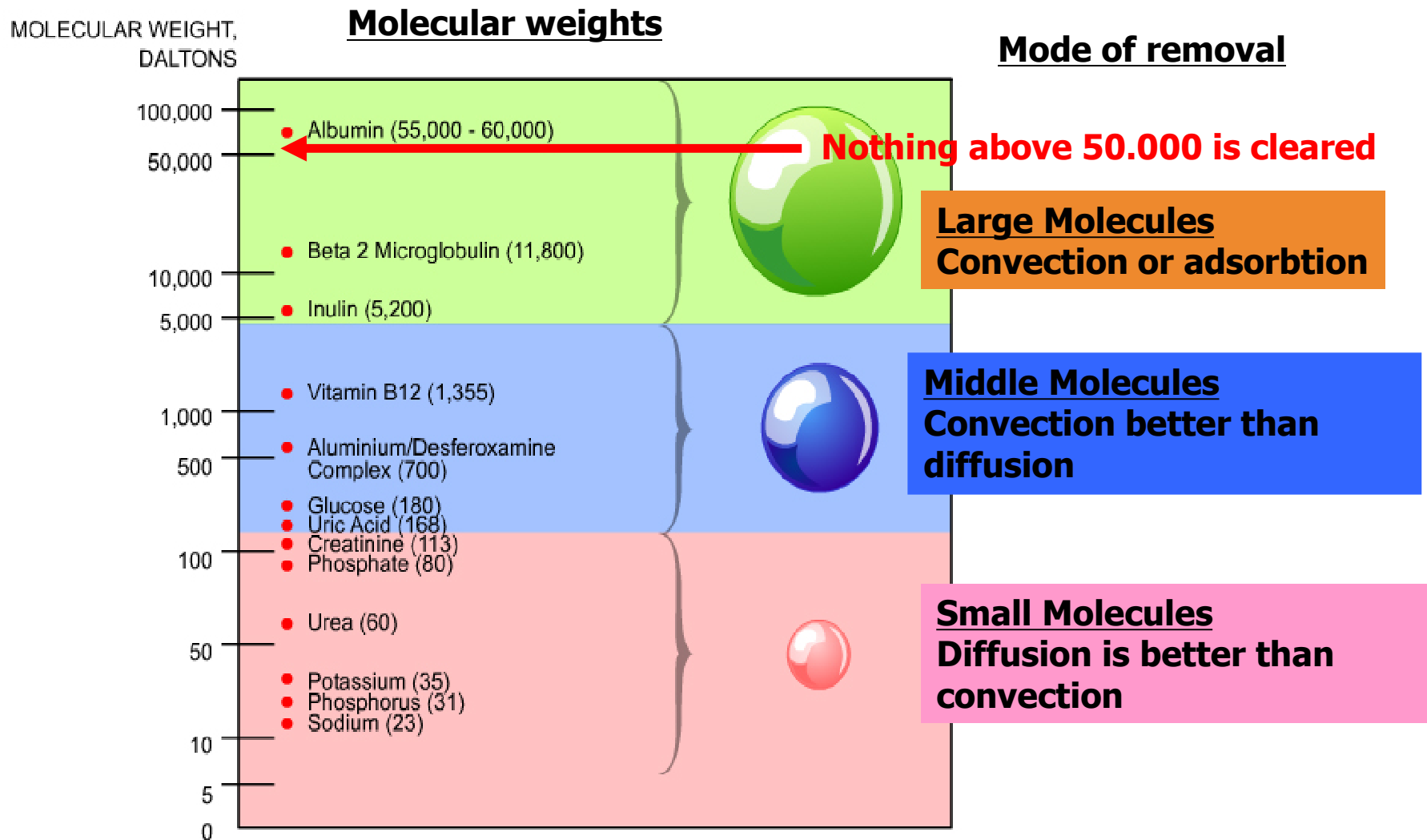
Solute drag depends on molecular weight and membrane characteristics

# Adsorption

Molecular adherence to surface or interior of a semi-permeable membrane.



# Size of molecules cleared by CRRT



# Continuous Renal Replacement Therapy

CRRT describes a group of Therapies.

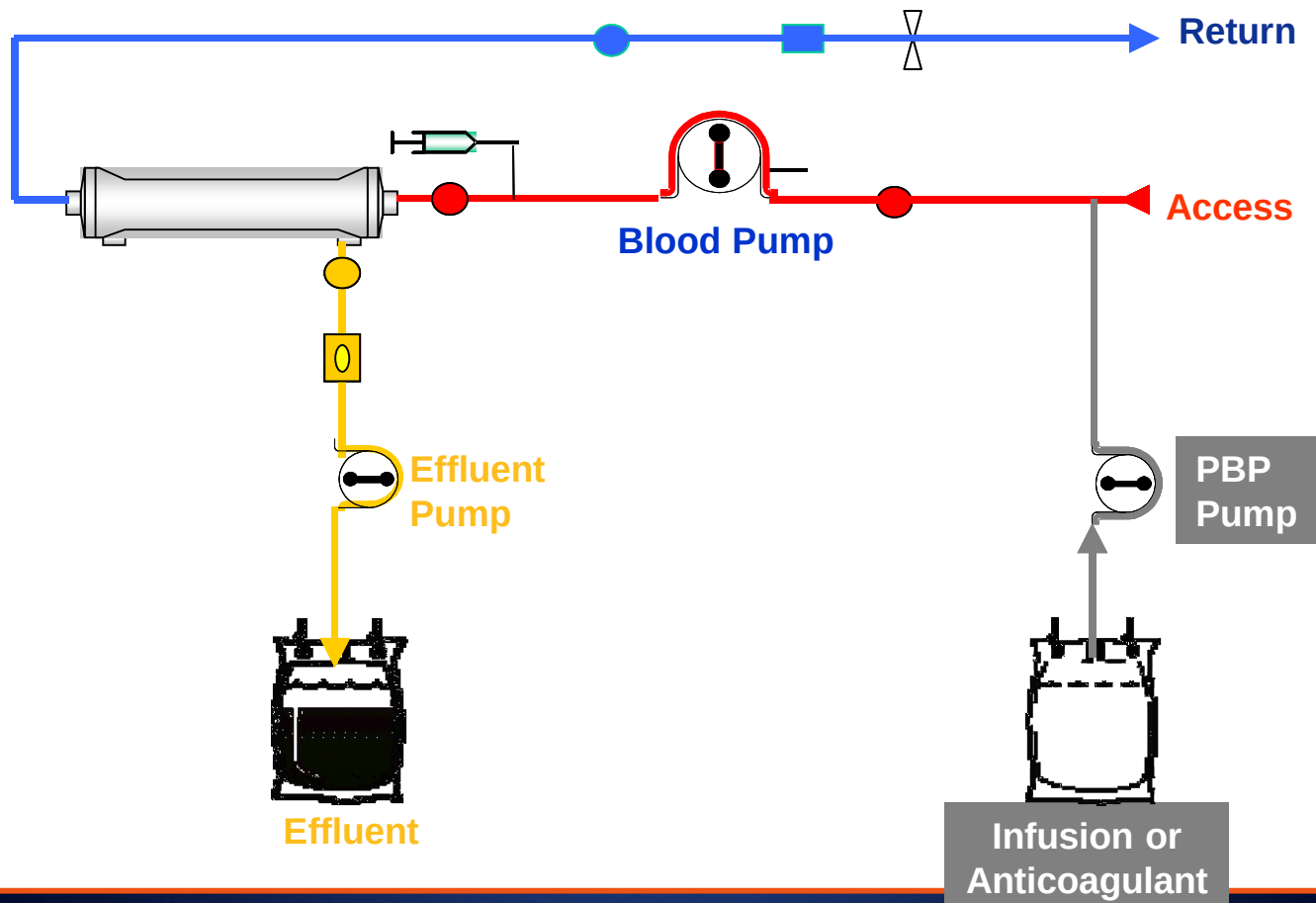
SCUF - Slow Continuous Ultrafiltration

CVVH - Continuous Veno-Venous Hemofiltration

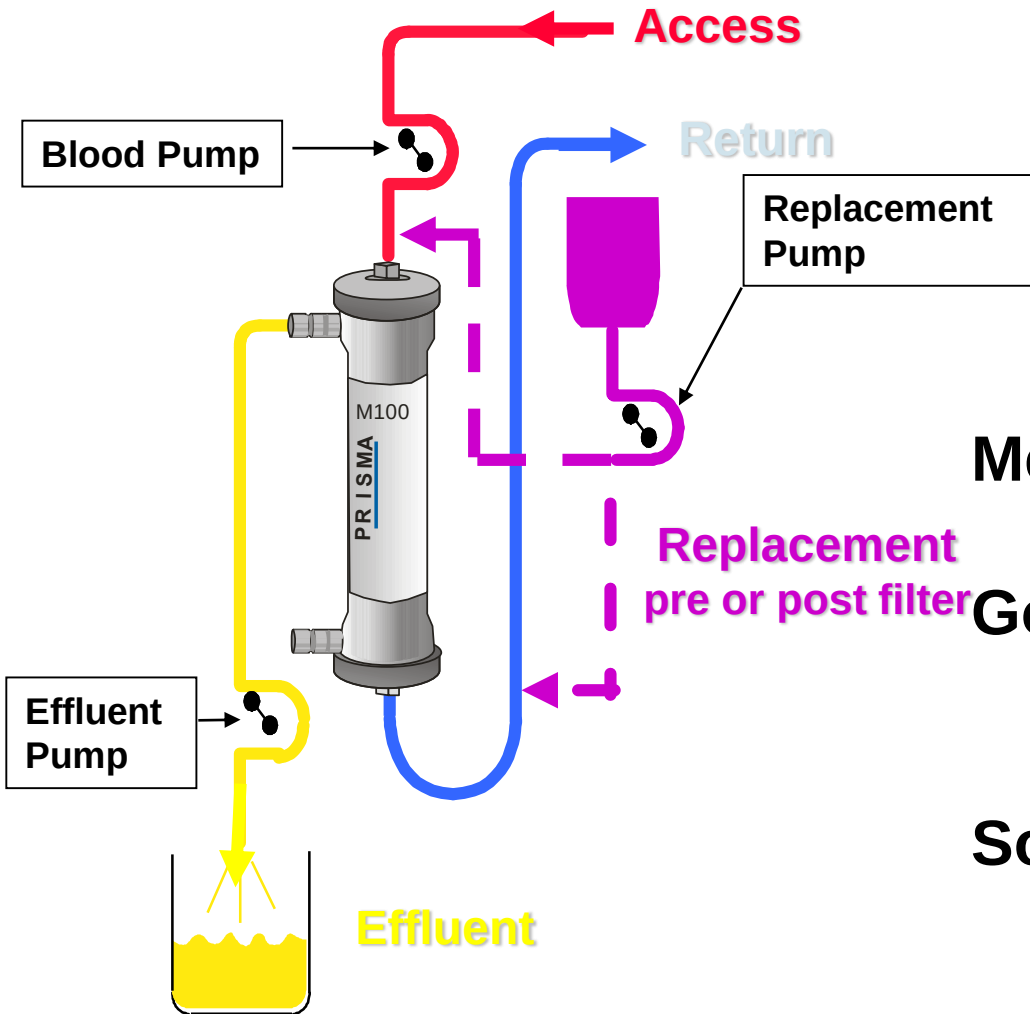
CVVHD - Continuous Veno-Venous Hemodialysis

CVVHDF - Continuous Veno-Venous Hemodiafiltration

# Slow Continuous UltraFiltration SCUF



# Continuous VV Hemofiltration



**Mechanism :** Convection,  
ultrafiltration

**Goals:** Safe fluid removal,  
removal of small -middle  
molecules

**Solution:** Replacement

# Pre vs Post Filter Dilution

## Pre - dilution

Dilutes blood before filter  
(Hct ↓)

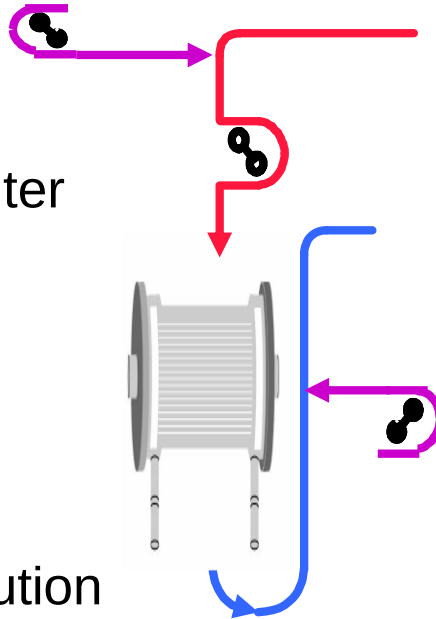
Reduces filter clotting

Prolonged filter life

Reduces effective  
clearance (appr. 15%)

More replacement solution  
required

Increase QB to overcome  
reduction in clearance



## Post - dilution

No reduction of clearance  
due to dilution of blood

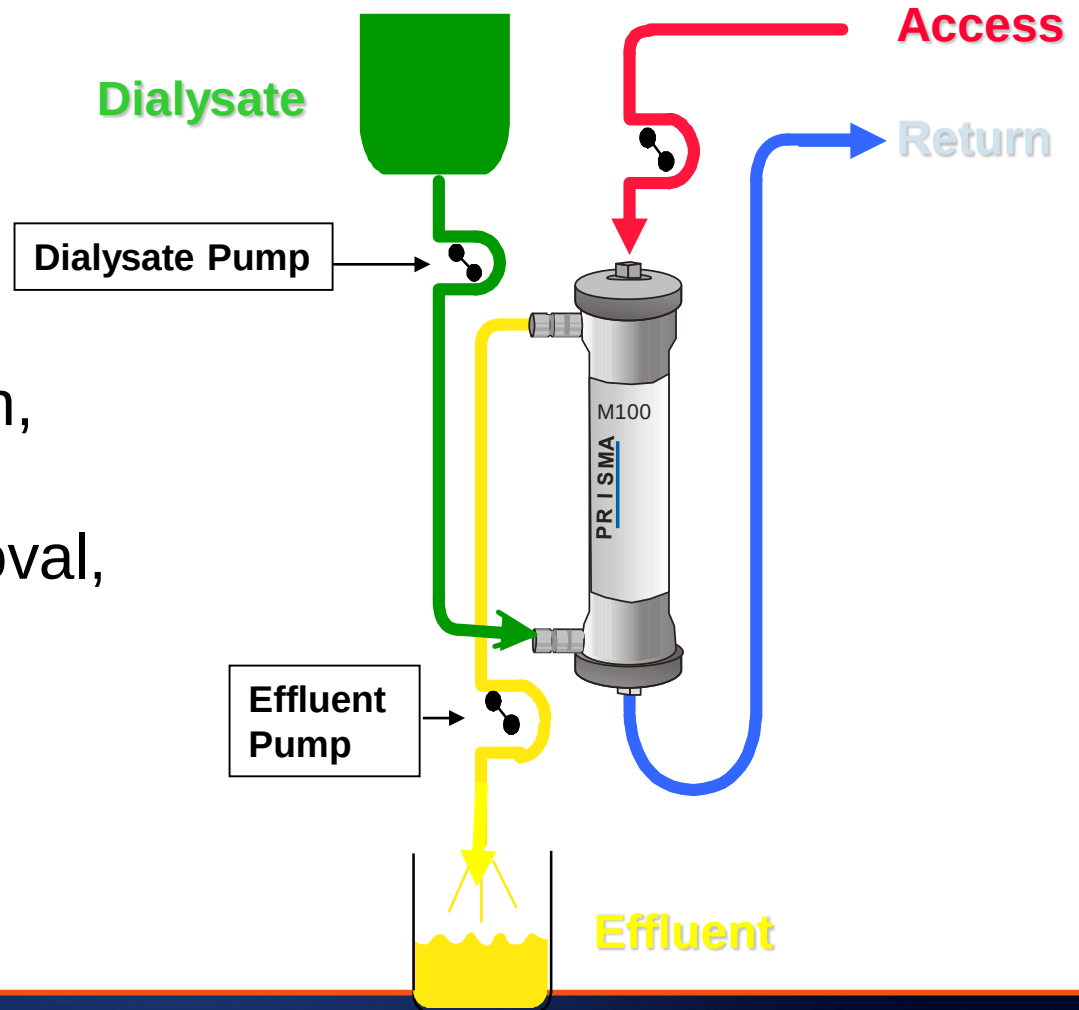
Less replacement solution  
required

Increased need of  
anticoagulant

Limits on UF rates due to  
hemoconcentration of  
blood

# Continuous VV HemoDialysis

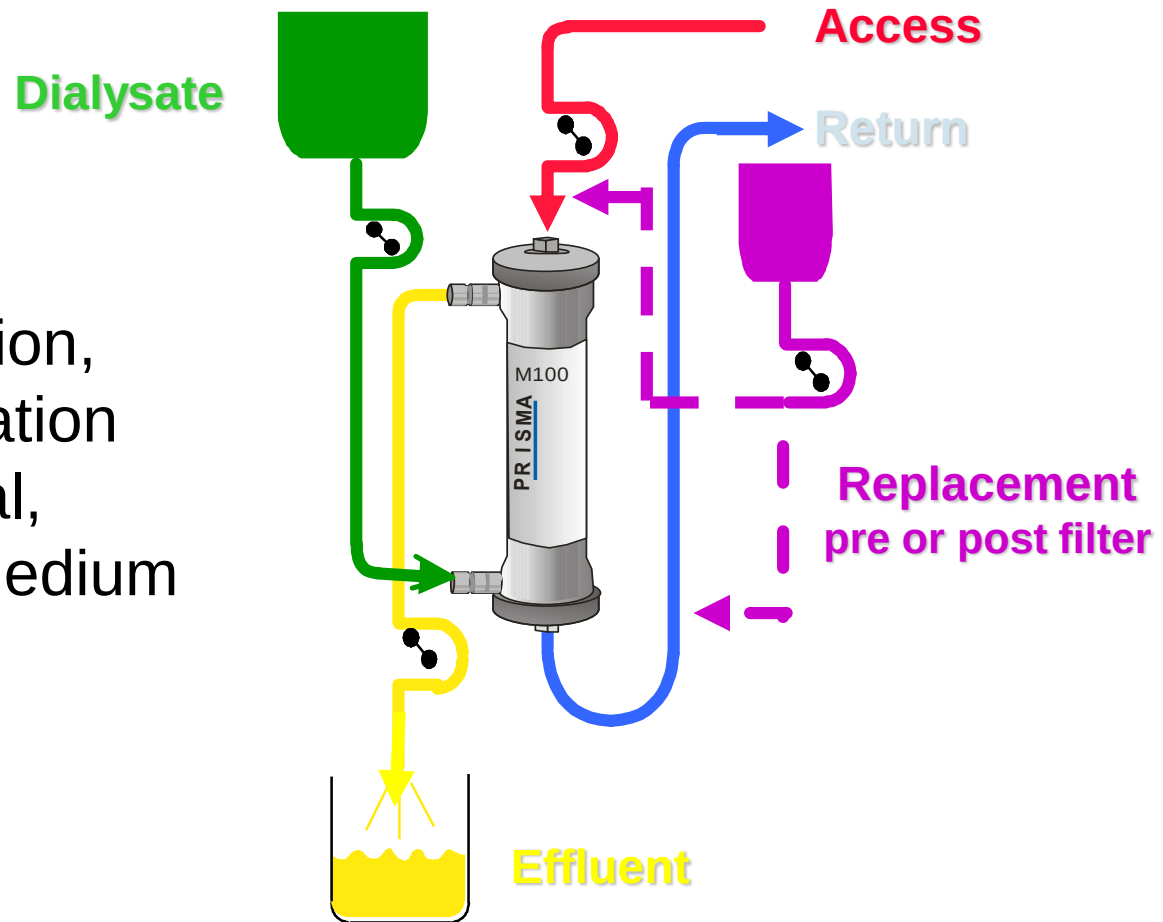
**Mechanism :** Diffusion,  
ultrafiltration  
**Goals:** Safe fluid removal,  
removal of small  
molecules  
**Solution:** Dialysate





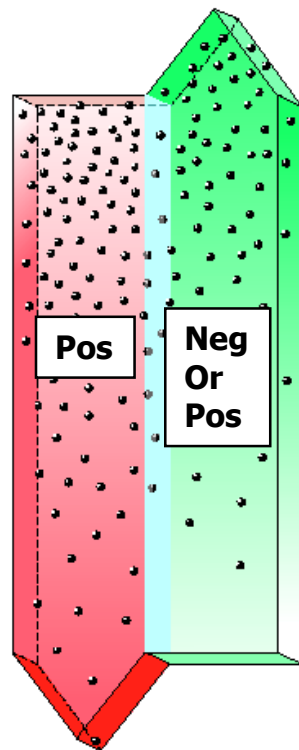
# Continuous VV HemoDiaFiltration

- **Mechanism** : Diffusion, convection, ultrafiltration
- **Goals**: Fluid removal, removal of small - medium molecules
- **Solution**: dialysate, replacement



# TMP Trans Membrane Pressure

Pressure difference between blood & fluid compartment



$$\text{TMP} = \frac{\text{Filter P} + \text{Return P}}{2} - \text{Effluent P}$$

Software calculated

A safety parameter, measuring function of filter

Maximum TMP ~ + 450 mmHg

Increased TMP → protein coating or clotting of fibers

# Replacement Solution

Fluid infused into the blood compartment pre or post filter

Contains electrolytes at physiological levels

Replaces the amount of fluid removed by the effluent or U.F pump

- does not affect the patient's fluid status

Drives convective transport

Facilitates the removal of both small and middle size solutes

# PBP Solution

Fluid infused into the blood compartment at the Access  
PBP pump can be used for

- Anticoagulation (citrate)
- Pre-dilution replacement
- Other solution

Total U.F = PBP + Replacement + pt. fluid removal rates  
PBP linked to blood pump

**Pre Blood Pump (PBP): Max 8L/hr**

# CRRT Therapy

Indications

Dosing

# Clinical Indications for CRRT

## Renal Indications

- **ARF** with oliguria or anuria
- **Azotemia** (a medical condition characterized by abnormal levels of urea, creatinine, various body waste compounds, and other nitrogen-rich compounds in the blood as a result of insufficient filtering of the blood by the kidneys)
- **Fluid overload**
- **Tumor lysis syndrome**
- **Sepsis**
- **Cerebral edema**

## Non-renal Indications

- **Drug overdose**
- **Metabolic disorders**
- **Crush injuries**
- **Sepsis**
- **ARDS**
- **Fluid overload**

# Role of CRRT in ARF

- Removal of toxins (urea, creatinine)
- Regulate electrolyte and water balance
- Regulate acid-base balance
- Prevent further damage to the kidney tissues
- Promote healing and renal recovery

# Role of CRRT in CHF

Maintain 24/7 fluid balance (All CRRT therapies)

- Reduce ascites and peripheral edema
- Relieve Pulmonary edema

Normalize cardiac filling pressures

Maintain 24/7 Electrolyte and acid/base balance (CVVHD, CVVHDF)

- - Dialytic control of electrolytes
- - Delivery of bicarbonate or lactate buffer

Possibly prevent further renal damage or prolong renal Insufficiency

Possibly reduce length of stay

Refractory Congestive Heart Failure: Overview and Application of Extracorporeal Ultrafiltration, Paul Blake and Emil P. Paganini, *Advances in RRT*, Vol 3, No2 (April), 1996: pp 166-173



# Role of CRRT in ARDS

- Maintain acid base balance
- Fluid control
- Regulate electrolyte balance
- Removal of toxins

# Role of CRRT in Sepsis

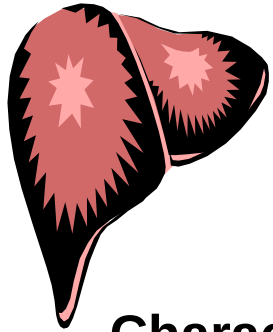
- Removal of middle to large molecule septic mediators by convection and adsorption including TNF- $\alpha$ , IL-1, IL-6 and IL-8 through:
- Removal of excess fluid and waste products
- Maintenance of acid-base balance
- Improve cardiovascular hemodynamic → removal of cardiodepressants (caused by inflammatory mediators)
- Thermoregulation

Bruce A. Molitoris. *Critical Care Nephrology* 2005. 28-34

# Role of CRRT for Rhabdomyolysis

- Maintain fluid, electrolyte, acid/base balance.
- Prevent further damage to kidney tissue
- Removal of small to middle protein molecules - 17,000 Dalton through convection
  - May cause nuisance BLD alarms





# *Hepatic Failure*

## **Characteristics**

Ascites

Coagulopathy

Elevated bilirubin

Hypomagnesemia

MOF

Inc. ICP/Cerebral edema

## **Role of CRRT**

Regulate and maintain fluid, electrolyte and acid-base balance \*

- Dec. ICP\*
- All CRRT

## **Remove toxins**

- excess bilirubin\*
- Inflammatory mediator

\*A new direction for dialysis. Sonia M. Astle, RN, MS, CCRN. RN Magazine, July 2001, pgs. 56-60

# *Intoxications*

Methanol, Ethylen-Glykol

Digoxin

Lithium

Valproat-Natrium

Phenobarbital

Lactate-acidosis (for example in case of Metformin)

# Role of CRRT in Intoxications

## Removal of small molecules by **Dialysis**

- Dialysate up to 4l/h
- Optional replacement up to 500ml/h

Duration: depends on clearance of toxic medication

# Determinants of Outcome

## Initiation of Therapy

- Rifle Criteria

## Dose

- Ronco, Bellomo Study

# Initiation of CRRT

## When do you start CRRT?

There is no clear consensus on when to start treatment. However, there are a few studies that favor earlier start to

in

Journal of Intensive Care Medicine (1997) 25, 305-314  
© Springer-Verlag New York, Inc. **CRRT in SALT**

L.G. Gettings  
H.N. Reynolds  
T. Scales

### **Outcome in post-traumatic acute renal failure when continuous renal replacement therapy is applied early vs. late**

“Earlier initiation of CRRT, based on pre-CRRT BUN, may improve the rate of survival of trauma patients who develop ARF.”

Dr. L.G. Gettings, Baltimore USA, 1989 to 1997

**Abstract:** The medical records of patients treated with CRRT for post-traumatic ARF were retrospectively reviewed. Chi-square testing was used to test frequencies between

Admission laboratory values including BUN, serum creatinine, lactate, and bicarbonate as well as fluid and blood requirements in the first 24 h were statistically the same. For the two groups, survival and mortality risk



# RIFLE Criteria

Established in 2002 ADQI conference, Vicenza, Italy

Presented in San Diego CRRT Conference in 2003

Purpose: To categorize patients based on renal function

Classification

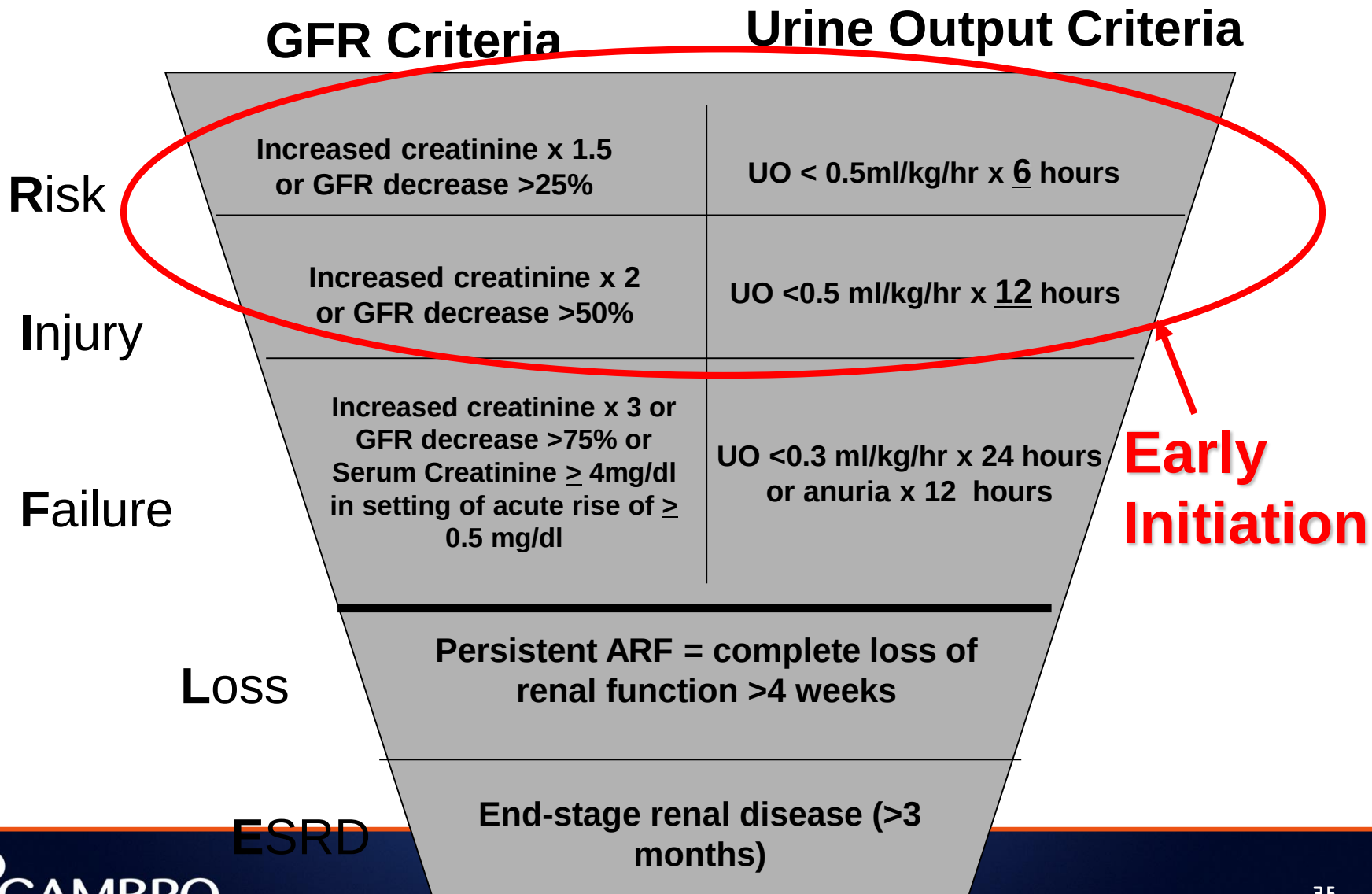
- Risk
- Injury
- Failure

Outcome:

- Loss
- End-stage renal disease

Bell, et. al., *Optimal follow-up after CRRT in ARF patients stratified with RIFLE criteria*. Nephrol Dialysis and Transplantation, 2005

# RIFLE Criteria



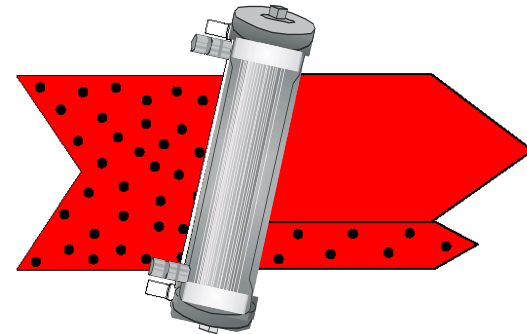
# CRRT Dose

## 1) Solute clearance – urea, creatinine

- $Kt/V = K \text{ (dialyzer clearance)} * t \text{ (treatment time)} / V \text{ (patient's volume)}$
- PRU (% reduction in urea)
  - In IHD, calculated as  $PRU = \frac{Pre - post}{pre}$
  - In CRRT,  $PRU = \frac{Access - return}{urea/creatinine}$
  - In CRRT,  $PRU = \frac{Effluent \text{ value in CVVHD, or post CVVH/DF}}{}$

## 2) Effluent flow rate rather than solute clearance

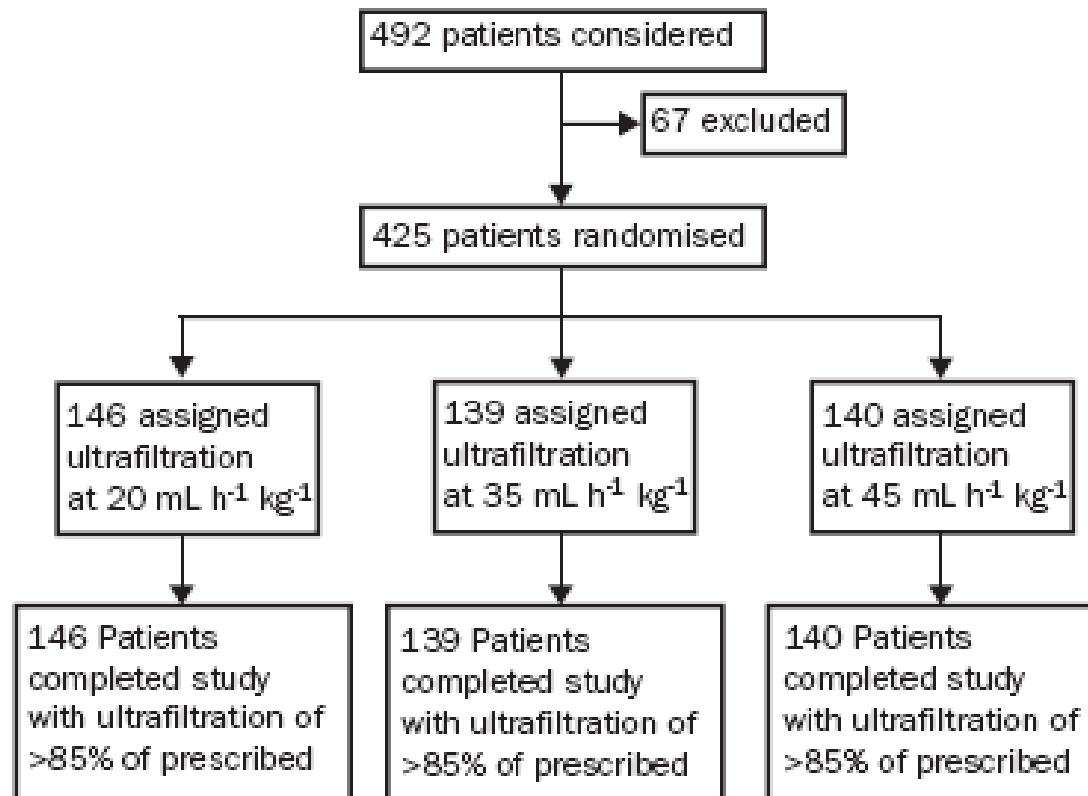
- Based on survival study by Ronco, Bellomo



Clearance (K) - volume of blood completely cleared of a substance/unit of time (ml/min).

# CRRT Dose

“What is an *adequate* dose for ARF?”



Ronco, Bellomo, et al. *Effects of different doses in CVVH on outcome of ARF*. Lancet, 2000

# CVVH Dose - Outcome

| Group | Dose<br>(ml/kg/hr) | Survival |
|-------|--------------------|----------|
| 1     | 20                 | 41%      |
| 2     | 35                 | 57%      |
| 3     | 45                 | 58%      |

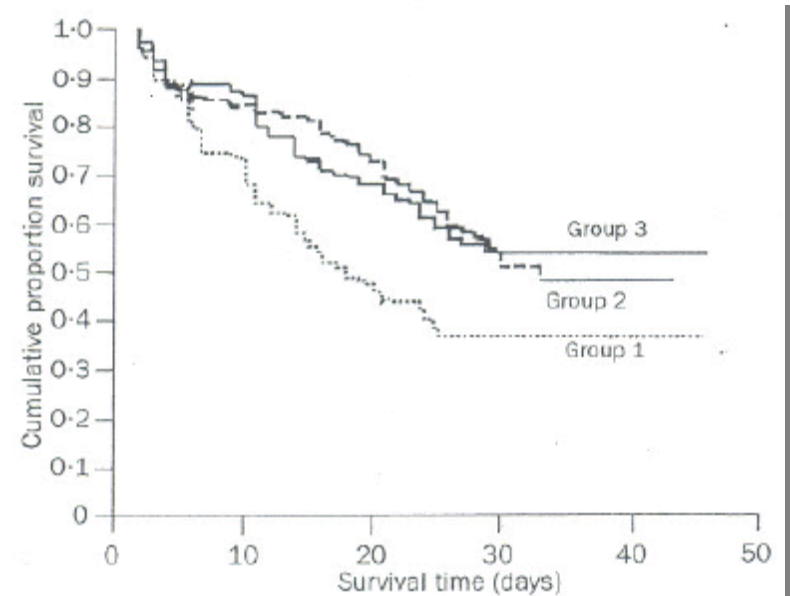


Figure 2: Kaplan Meier estimation of survival rates in the three groups

# CVVH Dose - Outcome

Treatment dose has an impact on mortality/ morbidity of ARF patients

- Start CVVH at **35 ml/h/kg** (eg. 70 kg patient = 2450 ml/h)

Continuous is more efficient and clinically tolerated than intermittent approaches

Early start has positive impact on outcome

Ronco, Bellomo, et. al.. *Effects of different doses in CVVH on outcome of ARF*. Lancet, 2000

# Therapy Table (Example)

|                                                                                | Acute Renal Failure                               | Sepsis                                             | Rhabdomyolyse                                    |
|--------------------------------------------------------------------------------|---------------------------------------------------|----------------------------------------------------|--------------------------------------------------|
| <b>Dose (ml/kg/BW/hr)</b>                                                      | <b>35</b>                                         | <b>50</b>                                          | <b>35</b>                                        |
| <b>Blood flow (ml/min)</b>                                                     | 150-350                                           | 250-450                                            | > 150                                            |
| <b>Patient 60 kg</b><br>Dialysate<br>Replacement Post<br>Replacement Pre = PBP | 60 x 35 = 2100 ml<br>900 ml<br>400 ml<br>800 ml   | 60 x 50 = 3000 ml<br>1200 ml<br>600 ml<br>1200 ml  | 60 x 35 = 2100 ml<br>500 ml<br>550 ml<br>1050 ml |
| <b>Patient 70 kg</b><br>Dialysate<br>Replacement Post<br>Replacement Pre = PBP | 70 x 35 = 2450 ml<br>1000 ml<br>450 ml<br>1000 ml | 70 x 50 = 3500 ml<br>1300 ml<br>700 ml<br>1400 ml  | 70 x 35 = 2450 ml<br>500 ml<br>650 ml<br>1300 ml |
| <b>Patient 80 kg</b><br>Dialysate<br>Replacement Post<br>Replacement Pre = PBP | 80 x 35 = 2800 ml<br>1000 ml<br>600 ml<br>1200 ml | 80 x 50 = 4000 ml<br>1400 ml<br>900 ml<br>1800 ml  | 80 x 35 = 2800<br>500 ml<br>750 ml<br>1550 ml    |
| <b>Patient 90 kg</b><br>Dialysate<br>Replacement Post<br>Replacement Pre = PBP | 90 x 35 = 3150 ml<br>1050 ml<br>700 ml<br>1400 ml | 90 x 50 = 4500 ml<br>1500 ml<br>1000 ml<br>2000 ml | 90 x 35 = 3150<br>500 ml<br>900 ml<br>1750 ml    |

# Exercise on Dosing

Mr. Smith, 60 kg, ARF

Required dose: 35ml/kg BW/hr

Mode: CVVHDF

- Pre: 66%
- Post: 33%
- Dialysate: 900ml/hr

**Calculation:  $60\text{kg} \times 35 \text{ ml/kg/h} = 2100 \text{ ml/h}$**

## Flow rates

900 ml Dialysate  
1200 ml Replacement  
2100 ml Total

A diagram with a central point from which two arrows branch out to the right. The top arrow points to '400 ml Post-Replacement' and the bottom arrow points to '800 ml Pre-Dilution (PBP)'.

400 ml Post-Replacement  
800 ml Pre-Dilution (PBP)



# Exercise on Dosing

Mrs. Jones, 100 kg, Polytrauma with Rhabdomyolysis

Required dose: 35ml/kg BW/hr

Mode: CVVH

- Pre-Replacement : 50%
- Post-Replacement :50%
- PBP: 500ml/hr

**Calculation:  $100\text{kg} \times 35 \text{ ml/kg/h} = 3500 \text{ ml/h}$**

## Flow rates:

3500 ml Total Replacement

```
graph LR; A[3500 ml Total Replacement] --> B[1500 ml Post-Replacement]; A --> C[1500 ml Pre-Replacement]; A --> D[500 ml PBP]
```

1500 ml Post-Replacement

1500 ml Pre-Replacement

500ml PBP

# Exercise on Dosing

Mr. Tan, 120 kg, ARF, pulmonary edema, and sepsis

Required dose: 45 ml/kg BW/hr

Mode: CVVHDF

- Pre- Replacement: 66%
- Post- Replacement: 33%
- Dialysate: 1200ml

**Calculation:  $120 \text{ kg} \times 45 \text{ ml/kg/h} = 5400 \text{ ml/h}$**

## Flowrates:

1200 ml Dialysate

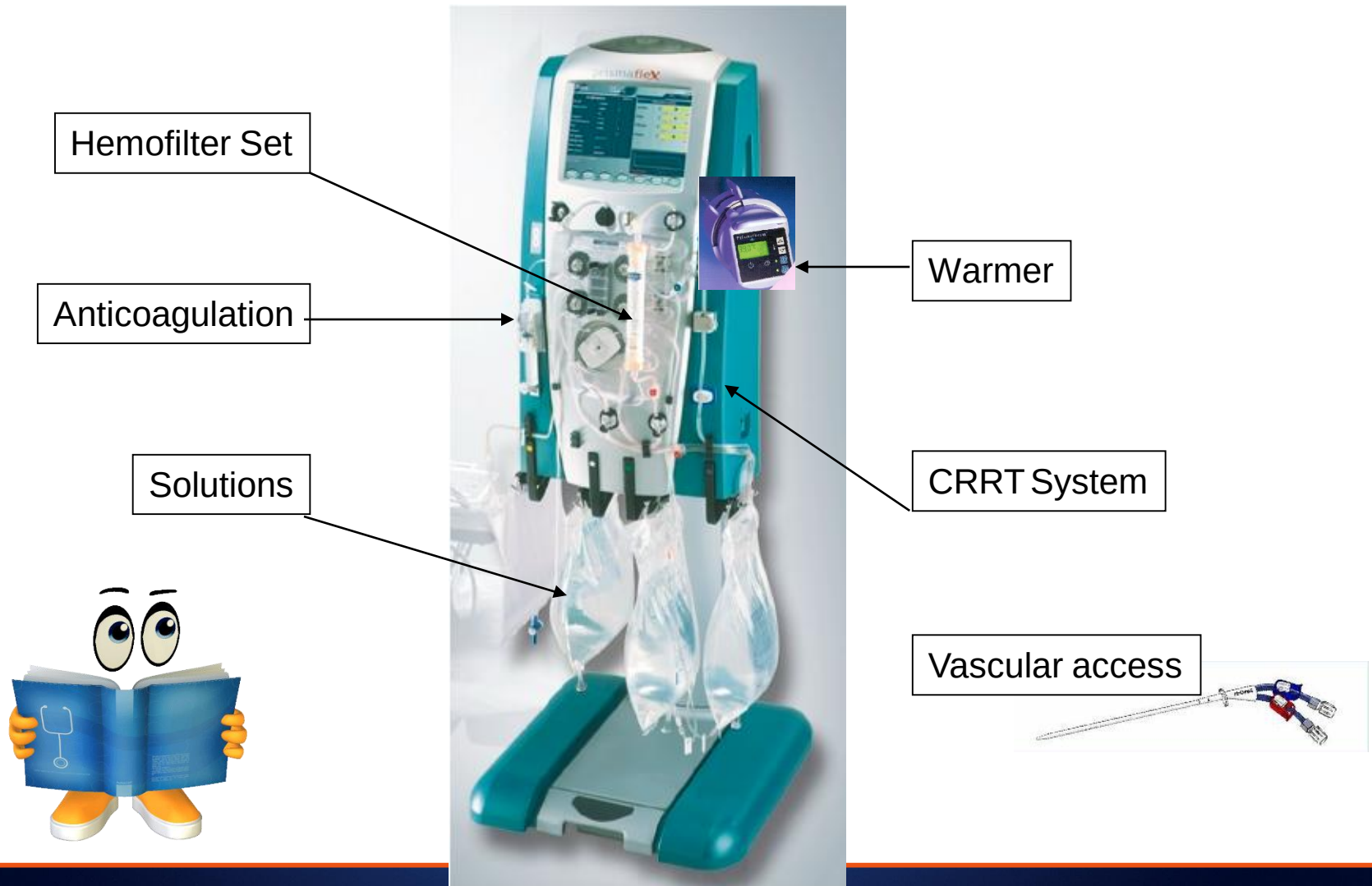
4200 ml Replacement

5400 ml Total

1400 ml Post-Dilution (Replacement)

2800 ml Pre-Dilution (PBP)

# Components of CRRT



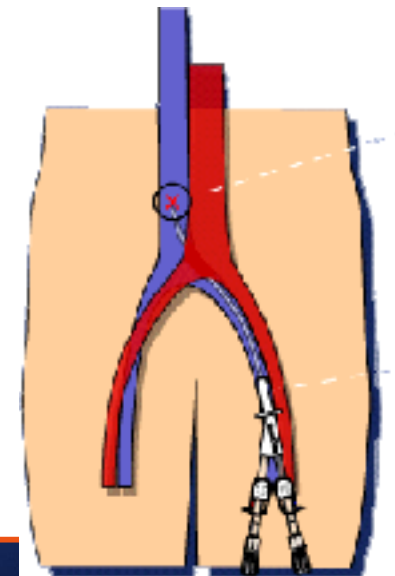
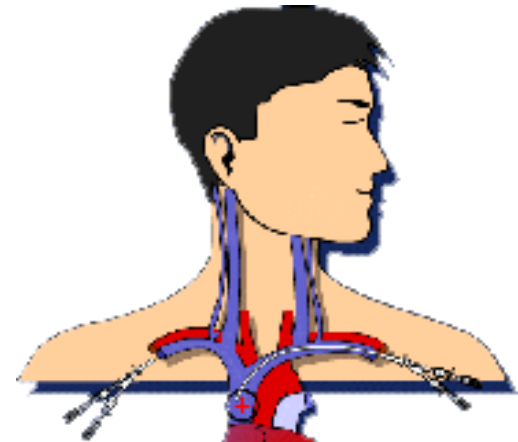
# Vascular Access

Types: double-lumen catheter

- Jugular, subclavian, or femoral

## Success Factors

- Proper size and placement
  - Right IJ: 15 cm
  - Femoral: 25 cm
- Proper assessment of patency



# Anticoagulation



Bleeding

Clotting

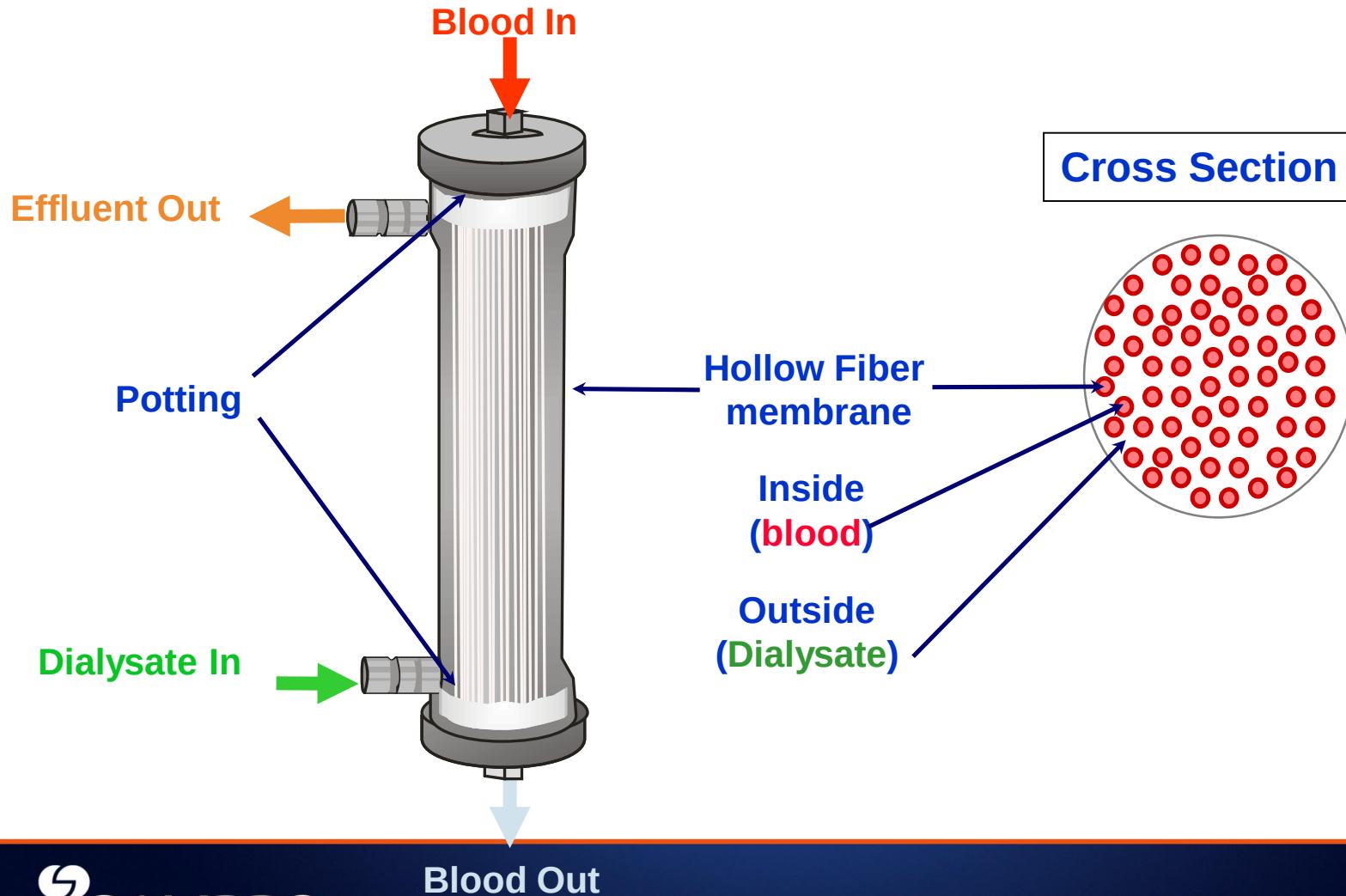
## Goals:

- Maintain patency of extracorporeal circuit
- Minimize patient complications of anticoagulation therapy

## Types:

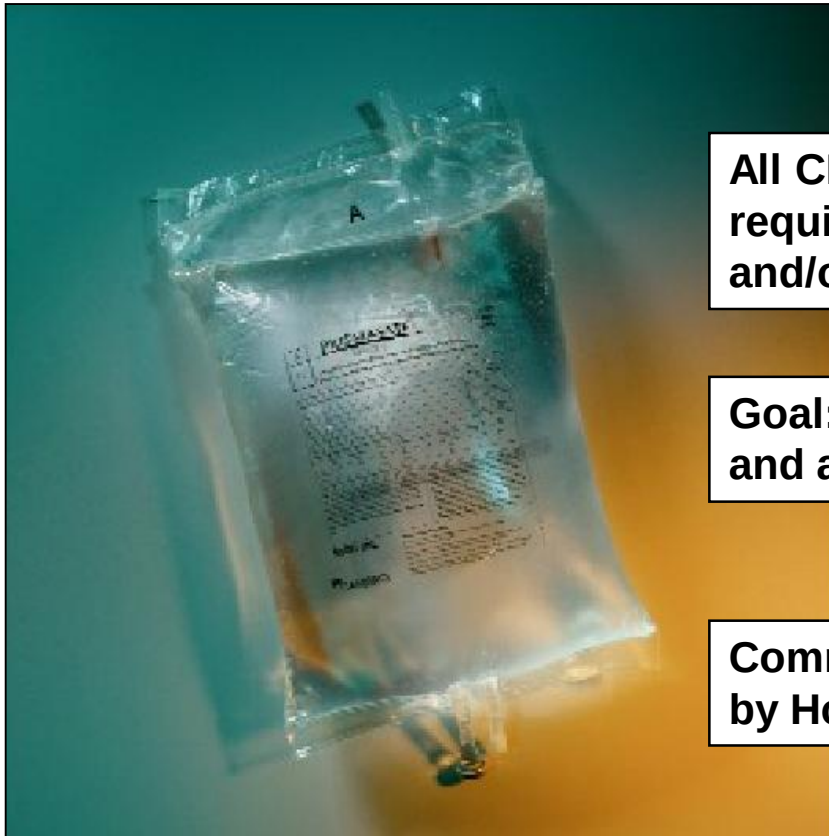
- Heparin
- Citrate

# The Basic Hemofilter



# Solutions in CRRT

Composition approximate to normal plasma water



**All CRRT techniques (except SCUF) require the use of sterile Dialysate and/or Replacement solution.**

**Goal: Correct electrolyte imbalance and achieve specific metabolic levels.**

**Commercially available or prepared by Hospital Pharmacy**

# Solution Buffers

## Lactate

Stable

Less costly

Requires metabolic activity to convert to bicarbonate

## Bicarbonate

More costly

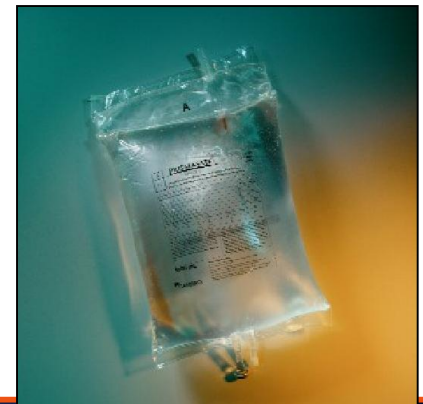
Less stable

More physiological and better tolerated

Improved control and correction of acidosis

### Gambro Bicarbonate Solutions

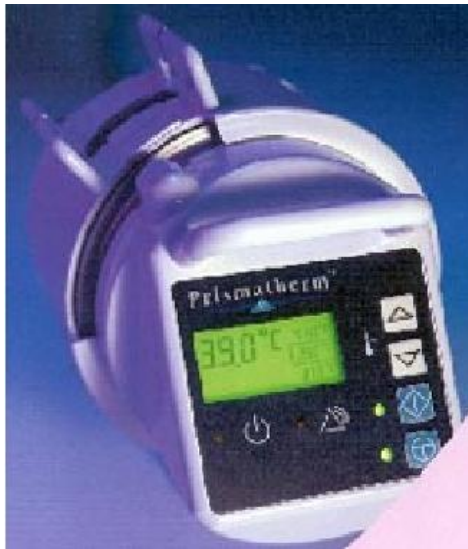
- two-compartment bag
- Fast, safe, and easy mixture of bicarbonate
- Prescription flexibility





# Warmers

Substantial heat loss in CRRT due to large fluid volume exchange  
Warmers prevents heat loss, maintain energy balance, and improve patient comfort



**Prismatherm II**  
**Blood warmer only**

